

Lab Assignment 5: Implementation of the Microservice Architecture

As part of the DMSA lab project, you will work on a continuous project in teams. Each project deals with a specific application scenario. We will guide you through the project step by step by means of lab assignments, which are published regularly.

In the fifth assignment, we implement our design as a microservice architecture with the help of Spring.

Due to the larger scope, a working period of *three weeks* is planned for this assignment (the next assignment is therefore not expected to appear until CW 26).

If you have any questions or uncertainties, please contact us (e.g. during the lab sessions)!

Task 1 (CW 23 - CW 24): Implementation of Bounded Contexts as Microservices

We use the approach discussed in the lecture materials and now implement each identified bounded context as a separate microservice. According to the division made in Assignment 4, each team member should therefore be responsible for exactly one microservice.

1. Realize the individual microservices with the technologies presented in the lecture, i.e. as *synchronous microservices* using Spring Boot, Spring MVC and REST (mandatory!). Take into account the basic requirements and technical constraints of your application scenario.
2. Define the interfaces required for the interaction between the individual microservices with the help of the context map developed in Assignment 3. *An integration of the microservices is not yet necessary in this task - this will take place in Task 2.* Any data and functions required from other microservices can initially be "mocked" with simple example data.
3. Implement a rudimentary user interface for *each* microservice so that the functionality (e.g. CRUD) can be easily tested and demonstrated. The technology for implementing

the user interface can be chosen freely. It makes sense to coordinate the realization of the user interfaces within the team so that they work as a meaningful overall application.

Important:

- Only deviate from the structure "1 Bounded Context $\Leftrightarrow$ 1 Microservice" in exceptional cases. Please discuss this with us in advance (e.g. as part of the lab sessions).
- If your design includes *fewer* bounded contexts than team members, divide the implementation work equally among them. Make sure that the division is traceable, e.g. via the commits in Git or alternatively via documentation of who has carried out which work.

Task 2 (CW 24 - CW 25): Integration of the Microservices

Please note: The lecture materials from CW 24 (learning modules *E-07* - *E-09*) are required to complete this task.

Now integrate the resulting microservices to form an overall system. Use at least the following patterns learned in the lecture for the integration:

1. Realize *service discovery* using Eureka.
2. Centralize your *configuration* with the help of Spring Cloud Config.
3. Add a *Circuit Breaker* with the help of Resilience4j.